Manual Statistics & Probability

SYNTHETIC DEMONSTRATION — NOT REAL COMPANY DATA

A1.1 Mean, median and mode

Data: 120, 150, 150, 180, 200, 220, 250, 250, 300, 380.

Mean: formula = sum(x)/n → substitution = (120+150+150+180+200+220+250+250+300+380)/10 = 2,200/10 = 220. Answer: 220. Interpretation: average basket is $220 in this toy example.

Median: ordered middle values are 200 and 220; formula = (x5+x6)/2 → (200+220)/2 = 210. Answer: 210. Interpretation: half of these observations are at or below $210.

Mode: 150 and 250 each occur twice; all others occur once. Answer: bimodal, 150 and 250. Interpretation: two repeated spending points appear in this small sample.

A1.2 Weighted mean

Formula = $\Sigma(w_i x_i)/\Sigma w_i$ → (0.10×100 + 0.20×150 + 0.30×220 + 0.40×300)/(0.10+0.20+0.30+0.40) = (10+30+66+120)/1.00 = 226.

Answer: 226. Interpretation: the weighted average order value is $226; higher weights on larger baskets raise the result.

A1.3 Median vs mean

Median is preferable when spending is skewed or contains a few unusually large orders. It describes a typical customer more robustly; mean remains useful for total-spend-per-customer planning.

A2.4 Range, quartiles and IQR

Ordered data: 10,12,14,15,18,20,22,25,30,45. Range = max−min = 45−10 = 35. Using the median-of-halves convention: lower half 10,12,14,15,18 gives Q1=14; upper half 20,22,25,30,45 gives Q3=25. IQR=Q3−Q1=25−14=11.

A2.5 1.5×IQR outlier rule

Lower fence = Q1−1.5×IQR =14−16.5=−2.5. Upper fence=Q3+1.5×IQR=25+16.5=41.5. Since 45>41.5, 45 is an outlier by this rule. Business meaning: investigate the unusually high value; do not automatically delete it.

A2.6 Sample variance and standard deviation

Data 8,10,12,14,16; n=5; mean=60/5=12. Sample variance $s^2=\Sigma(x-\bar{x})^2/(n-1)=[16+4+0+4+16]/4=40/4=10$. Sample SD $s=\sqrt{10}=3.1623$. Answer: variance 10 units^2; SD≈3.16 units.

A2.7 High standard deviation

High SD means observations vary widely around the mean. For spending it can indicate heterogeneous customer behavior and make averages less representative; segmenting or using robust statistics may be more useful.

A3.8 Z-score

$z=(x-\mu)/\sigma=(85-70)/10=15/10=1.5$. Answer: 1.5 SD above the mean.

A3.9 Compare unusual observations

For 660: $z=(660-500)/80=160/80=2$. For 350: $z=(350-500)/80=-150/80=-1.875$. Compare $|z|$: $2>1.875$, so 660 is more unusual relative to this normal-scale reference.

A3.10 Histogram and box plot

Histogram shows distribution shape, peaks, skew and gaps. Box plot summarizes median, quartiles and IQR and flags potential outliers, making group comparisons compact.

A4.11 Payment probabilities

Online-paid=600/1000=0.60=60%; COD=250/1000=0.25=25%; wallet=150/1000=0.15=15%. They sum to 1.00.

A4.12 Empirical return probability

P(return)=observed returns/observed orders=120/1000=0.12=12%. This estimates the rate under the sample period and process.

A4.13 Three probability types

Classical: if each of four equally likely promo cards is equally likely to be drawn, P(one chosen card)=1/4. Empirical: return probability estimated as historical returns/orders. Subjective: a merchandiser's forecast that a new product will sell through, based on experience and incomplete evidence.

A5.14 Conditional late delivery among churned

$P(\text{Late}|\text{Churn})=180/300=0.60=60\%$.

A5.15 Bayes: churn among late deliveries

$P(\text{Churn})=300/2000=0.15$; $P(\text{Late}|\text{Churn})=180/300=0.60$; $P(\text{Late}|\text{Not churn})=340/1700=0.20$. Bayes: $P(\text{Churn}|\text{Late})=P(\text{Late}|\text{Churn})P(\text{Churn})/P(\text{Late})$, where $P(\text{Late})=(180+340)/2000=0.26$. Substitution$=(0.60\times0.15)/0.26=0.09/0.26=0.34615\approx34.6\%$. Count check: $180/(180+340)=34.6\%$.

A5.16 Conditional probability direction

$P(A|B)$ conditions the denominator on B; $P(B|A)$ conditions it on A. Their denominators/base rates differ, so they are generally unequal.

A6.17 Binomial exactly two returns

For n=5, p=.20, k=2: $P(X=2)=C(5,2)(.2)^2(.8)^3=10\times.04\times.512=.2048=20.48\%$, assuming independent orders with constant p.

A6.18 Poisson exactly two tickets

$\lambda=3$ tickets/hour. $P(X=2)=e^{(-3)}3^2/2!=e^{(-3)}\times4.5\approx0.2240=22.40\%$, assuming a stable rate and independent arrivals.

A6.19 Empirical rule

For an approximately normal distribution, about 68%, 95%, and 99.7% of observations lie within 1, 2, and 3 SDs of the mean, respectively. It is a rule of thumb, not universal.

A7.20 Hypotheses

H0: $\mu_{treatment}-\mu_{control}\leq0$ (new checkout does not increase AOV). H1: $\mu_{treatment}-\mu_{control}>0$ (it increases AOV). Define the population, exposure, and measurement window before testing.

A7.21 Type I and II errors

Type I: conclude AOV increased when it did not (false positive), potentially rolling out an ineffective change. Type II: miss a real increase (false negative), potentially foregoing value.

A7.22 p-value and tails

A p-value is the probability, assuming H0, of data at least as incompatible with H0 as observed. It is not the probability H0 is true. One-tailed tests a pre-specified direction; two-tailed tests either direction.

A7.23 Test selection

Two-sample t-test: compare two independent group means (or paired t-test for paired observations), subject to assumptions/robust alternatives. ANOVA: compare means across 3+ groups. Chi-square: association between categorical variables or goodness-of-fit.

A8.24 A/B groups

Control: eligible customers see existing checkout. Treatment: eligible customers see new checkout. Randomize at customer level and avoid cross-device contamination where feasible.

A8.25 Primary and guardrail metrics

Primary: revenue per eligible randomized customer (or conversion if pre-registered). Guardrail: checkout error rate or refund rate. Choose one primary metric and define attribution window in advance.

A8.26 Sample size and MDE

Sample size determines precision/power. Minimum detectable effect (MDE) is the smallest business-relevant effect the design aims to detect. Too-small samples can miss real effects; repeated peeking/multiple metrics inflate false positives unless controlled.

Validation and scope

Manual calculations above use assessment-provided values. The synthetic company dataset is separate and is used only for the portfolio analysis.